

Beyond Classical Computer Science: Educating for Quantum Computing in Colombia

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Abstract. Quantum computing has moved from a physics curiosity to an emerging engineering discipline, and with that shift comes a workforce question that most computing programs are not equipped to answer. This review argues that traditional computer-science education, and, in the Colombian case, the *Ingeniería de Sistemas* tradition that stands in for it, does not by itself produce practitioners able to understand, design, and build quantum-computing solutions. The gap is structural rather than incremental: the competencies that make quantum computing *quantum* (superposition and measurement, complex Hilbert-space reasoning, the circuit model, interference-based algorithm design, and noise/error correction) live almost entirely outside the classical computing curriculum, while the software, systems, and application competencies needed to turn quantum devices into useful tools live almost entirely outside physics. Colombia faces this gap under particular pressure: computer science as a named discipline is young here, subsumed for decades within *Ingeniería de Sistemas*, while quantum knowledge has been confined to physics and chemistry departments. We frame the problem against the international literature on quantum workforce and education, compare three Latin American national visions (Colombia's Colombian Quantum Alliance, Chile's 2025–2035 national strategy, and Mexico's IMTC initiative), and propose a concrete, phased teaching pathway aligned to the CQA's 2026–2036 roadmap. Two exhibits anchor the argument: a competency-coverage matrix across computer science, physics, and *Ingeniería de Sistemas*, and a layered education pathway keyed to the CQA phases.

Keywords: Quantum computing education · Computer science curricula · Quantum workforce · Colombia · *Ingeniería de Sistemas* · Latin America

1. Introduction: a second quantum revolution needs a new kind of practitioner

The technologies now grouped under the "quantum" label — computing, communication, and sensing — rest on deliberately engineering, rather than merely observing, individual quantum systems. Dowling and Milburn named this shift the "second quantum revolution," distinguishing the passive use of quantum mechanics (transistors, lasers) from the active control of superposition and entanglement as engineering resources [1]. Quantum computing is the most demanding expression of that shift. Its theoretical promise is concrete and old enough to be well established: Shor's factoring algorithm showed that a quantum computer could break widely used public-key cryptography in polynomial time [2], and Grover's search algorithm gave a provable quadratic speedup for unstructured search [3]. The canonical synthesis of the field's mathematical and computational foundations remains Nielsen and Chuang's text [4].

What has changed in the last decade is that the hardware caught up enough to make the workforce question urgent. Programmable superconducting processors have executed sampling tasks beyond the reach of classical simulation [5], and the field has settled into what Preskill called the "noisy intermediate-scale quantum" (NISQ) era — machines large enough to be interesting, too noisy to be fault-tolerant, and useful only to practitioners who understand their limits [6]. Building anything real on these devices demands a blend of skills — physics intuition, linear-algebra fluency, algorithmic thinking, software engineering, and hardware-aware error management — that no single traditional degree assembles.

This review is about that assembly problem, and specifically about how a country like Colombia, where the computing discipline is itself young and quantum knowledge sits inside physics, should build the educational pathway to solve it. We take as our reference points the recent analysis of Colombia's quantum ecosystem [7] and the vision articulated by the Colombian Quantum Alliance [8], and we read them against the international education literature and two neighbouring national strategies.

2. The disciplinary gap: why classical computer science is not enough

2.1 The argument in one figure

The central claim of this review is that quantum-computing competence is not an extension of computer science, physics, or systems engineering — it is an intersection that none of them covers on its own. To make that concrete, we assembled a coverage matrix of fifteen competencies that a working quantum-computing practitioner needs, and

asked, for each, whether a typical undergraduate program of each type treats it as **core**, **partial/optional**, or **absent**.

No single discipline covers the quantum-computing competency set
Coverage in a typical undergraduate program of each type; six quantum-specific rows are absent from both Computer Science and Ingeniería de Sistemas

Linear algebra over $\mathbb{C}$ (complex vector spaces, Hilbert spaces, tensor products)	partial	core	partial
Probability & statistics (measurement, expectation, sampling)	core	core	partial
Quantum mechanics postulates (superposition, measurement, unitarity)	absent	core	absent
Discrete math, logic & complexity theory	core	partial	partial
Algorithms & computational thinking	core	partial	core
Quantum information theory (qubits, no-cloning, entropy, channels)	absent	partial	absent
Quantum circuit model & gate-level reasoning	absent	partial	absent
Quantum algorithms (Shor, Grover, VQE/QAOA, HHL)	absent	partial	absent
Reversible & amplitude-based reasoning (interference, phase)	absent	partial	absent
Programming & software engineering	core	partial	core
Numerical methods, simulation & HPC	partial	partial	partial
Error models, noise & quantum error correction	absent	partial	absent
Cloud & platform engineering (access to remote QPUs)	partial	absent	partial
Domain modelling for use-inspired applications (optimization, chemistry, ML)	partial	partial	partial
Post-quantum cryptography & security	partial	absent	partial
	Computer Science	Physics	Ingeniería de Sistemas

Fig. 1. Competency-coverage matrix across Computer Science, Physics, and Ingeniería de Sistemas. Six quantum-specific competencies are absent from both the CS and IS curricula and only partial in physics.

The pattern in the figure is the argument. Computer science owns the software and algorithmic core, programming, discrete mathematics, complexity theory, computational thinking, but treats every quantum-specific competency as absent: the quantum-mechanical postulates, quantum information theory, the circuit model, quantum algorithms, interference-based reasoning, and error correction are simply not in the curriculum. Physics is the mirror image: it covers the quantum substrate well but treats software engineering, cloud/platform engineering, and post-quantum cryptography as peripheral or absent. *Ingeniería de Sistemas*, the Colombian default, is the weakest of the three on the quantum-specific rows and only partial on several of the mathematical foundations.

2.2 Why the quantum rows do not reduce to classical training

It is tempting to assume that a strong programmer can "pick up" quantum computing the way one picks up a new framework. The literature on quantum education argues otherwise, and the reason is conceptual, not merely a matter of syllabus coverage. Quantum computing rests on a small set of ideas — superposition, measurement as a probabilistic and state-collapsing operation, entanglement, and the manipulation of complex probability *amplitudes* — that have no classical analogue and that students consistently find counterintuitive. Fox, Zwickl, and Lewandowski, surveying the role of higher education in preparing for the quantum revolution, found that programs must deliberately teach these concepts and the mathematical language (complex linear algebra over Hilbert spaces) that expresses them, rather than assuming transfer from classical courses [9].

The distinctive competency is what we labelled *amplitude- and interference-based reasoning*. Classical algorithm design reasons about states and transitions; quantum algorithm design reasons about engineering the constructive and destructive interference of amplitudes so that wrong answers cancel and right answers reinforce. Shor's and Grover's algorithms are unintelligible without it, and it is precisely the mode of thinking that a classical CS curriculum never exercises. This is why efforts to introduce quantum computing to engineering and computer-science students treat the linear-algebra-plus-interference substrate as the hard prerequisite, not the programming [10, 11].

2.3 But physics alone is not enough either

The gap runs the other way too, and this is the point most often missed when quantum computing is treated as "a physics topic." Turning a quantum device into a solution requires software engineering (building, testing, and versioning quantum programs), platform engineering (accessing remote QPUs through cloud services), systems thinking about hybrid classical–quantum workflows, and a bridge to an application domain — chemistry, optimization, machine learning, cryptography. These are the competencies the matrix shows as weak or absent in a

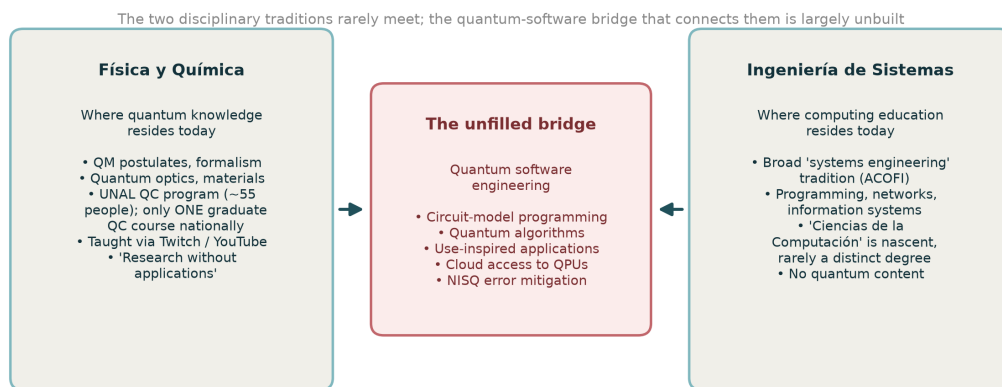
physics degree. The materials and hardware challenges that dominate the physics research frontier [12] are largely orthogonal to what an *applications* practitioner needs. A quantum internet or a fault-tolerant machine will be built by physicists and materials scientists; a quantum *solution* to an Andean energy-grid or a tropical-biodiversity problem will be built by someone who can also write and deploy software and talk to a domain expert.

The empirical education literature converges on the same conclusion from the classroom side. Meyer and colleagues, surveying today's quantum-information courses, describe them as inherently interdisciplinary spaces drawing students from physics, computer science, and engineering simultaneously, and note the pedagogical difficulty of serving audiences with such different preparation [13]. Expert-panel work on quantum-information-science education reaches a similar verdict, emphasizing that programs must intentionally build bridges across the contributing disciplines rather than bolt quantum content onto an existing major [14]. The conclusion for curriculum design is unambiguous: quantum-computing education is a co-teaching problem, not a single-department problem.

3. The Colombian specificity: computer science as a young discipline, quantum inside physics

The disciplinary gap described above is universal. What makes it acute in Colombia is that the "classical" side of the gap is itself underdeveloped and structurally displaced, and the quantum side is confined to a single research niche. Two peculiarities of the Colombian academic landscape combine to widen the gap.

Colombia's structural gap: quantum sits inside physics, computing sits inside Ingeniería de Sistemas



Sources: Bello et al. 2024 (arXiv:2409.00059); ACOFI systems-engineering curriculum; UNAL Ingeniería de Sistemas y Computación

Fig. 2. Colombia's structural gap: quantum knowledge resides in physics and chemistry, computing education resides in Ingeniería de Sistemas, and the quantum-software bridge between them is largely unbuilt.

3.1 Computing is *Ingeniería de Sistemas*, and computer science is nascent

In most of Colombian higher education, the undergraduate degree that trains people to work with computers is not "Ciencias de la Computación" but "Ingeniería de Sistemas" — a broad systems-engineering tradition oriented toward information systems, networks, software development, and the management of technology in organizations. The reference curriculum maintained by the Colombian association of engineering faculties (ACOFI) frames the field in these systems-engineering terms rather than as computer science in the ACM/IEEE sense [15]. Even the country's flagship program at the Universidad Nacional carries the composite name "Ingeniería de Sistemas y Computación," signalling that computation is an aspect of systems engineering rather than a discipline in its own right [16]. The disciplinary history has been retrospectively examined within Colombia itself, tracing how the systems-engineering label came to dominate and what it left out [17].

This is not a merely nominal distinction. A widely-circulated 2015 critique argued bluntly that essentially no Colombian university taught software engineering, or computer science, as a rigorous discipline in its own right, and that the systems-engineering framing diluted the theoretical and mathematical core [18]. Whatever one makes of the

polemic, its premise matters for quantum computing: the theoretical foundations that quantum computing leans on most heavily — complex linear algebra, discrete mathematics, complexity theory, and the habit of reasoning about computation abstractly — are exactly the parts of a computer-science education that a systems-engineering framing tends to underweight. The matrix in §2 marks several of these as only "partial" for *Ingeniería de Sistemas* for this reason.

3.2 Quantum knowledge is confined to physics, and thin even there

On the other side of the gap, quantum expertise in Colombia sits almost entirely within physics and chemistry departments, and the intersection with computing is genuinely new. The most detailed account of the national ecosystem, a triple-helix analysis of academia, industry, and government, found that Colombia had essentially no quantum investment or legislation, and that quantum-computing teaching was concentrated in a single university program [7]. That study characterized the Universidad Nacional's quantum-computing program — roughly fifty-five professors and students, as offering the only graduate-level quantum-computing course in the country, disseminating much of its teaching through public streaming channels, and being, in the authors' framing, "research without applications." On the industry side it documented a single prominent application, a consumer-goods company using quantum optimization software for a product-formulation problem, which it characterized as "applications without buyers." The picture is of two small islands, an academic one and an industrial one, with almost no educational bridge between them and no policy scaffolding around them.

3.3 Why this raises the stakes

Put the two peculiarities together and the Colombian situation is sharper than the generic disciplinary gap. In a country with mature computer-science and physics departments, building quantum-computing education means connecting two strong traditions. In Colombia it means connecting a *displaced* computing tradition (strong in systems and software practice, thinner in the theoretical foundations quantum computing needs) with a *narrow* quantum tradition (concentrated in one program, oriented to research rather than application). The bridge in the middle of the figure, quantum software engineering, use-inspired applications, cloud access to real hardware, is not merely underdeveloped; it has almost no natural institutional home in the current structure. That is the problem any Colombian quantum-education strategy has to solve.

4. The Colombian Quantum Alliance vision as organizing frame

The Colombian Quantum Alliance (CQA), convened by CyberColombia, offers a strategy that is, in effect, a direct response to the structural gap of §3 [8]. Its defining choice is what it calls a "Leapfrog Strategy": rather than competing on quantum *hardware* — which demands capital, cleanroom fabrication, and a materials-science base Colombia does not have — the Alliance positions the country to compete on the *application and algorithm layer*, an "Algorithm Factory" model that is asset-light and software-first. Hardware is accessed, not owned: the CQA frames CyberColombia as an orchestrator, bridge, and catalyst that reaches real quantum processors through cloud and high-performance-computing partnerships (for example with NCSA and the regional SCALAC network) rather than building its own.

Two features of this vision make it a useful organizing frame for education. First, its research philosophy is explicitly *use-inspired* — Pasteur's-quadrant work that pursues fundamental understanding and concrete national applications at once, with candidate domains such as tropical-biodiversity mapping and Andean energy-grid optimization. That immediately implies an educational requirement: practitioners who can bridge quantum methods to a problem domain, exactly the "domain modelling" competency the matrix marks as only partial everywhere. Second, the CQA embeds education in a staged, decade-long roadmap with measurable targets rather than treating it as an afterthought. The roadmap runs in three phases: **Phase I, Foundation & Literacy (2026–2028)**, centred on training the trainers, standing up a Quantum Cloud Gateway, and enrolling a first quantum master's cohort, with a target of 100+ students and 10+ specialized professors; **Phase II, Acceleration (2029–2031)**, adding an "Andean Quantum Sandbox," post-quantum-cryptography advisory work, and company-funded proofs of concept; and **Phase III, Scaling & Sovereignty (2032–2036)**, targeting startups, a hosted regional summit, a workforce of 500+, and the first sovereign patents. The Alliance also situates itself regionally through a "Quantum Latino" integration agenda.

The strategic logic is coherent and, importantly, matched to the constraint diagnosed in §3: a software-first, application-layer bet is exactly the bet a country with a strong software-practice tradition and a thin hardware base *should* make. But it also sharpens the educational challenge rather than dissolving it. An Algorithm Factory needs

people who can reason about quantum algorithms and interference (absent from the CS and IS curricula), write and deploy robust quantum software (absent from physics), and connect both to a domain — the full width of the matrix, delivered by co-teaching. The rest of this review asks how to build that, drawing first on international experience and then on two neighbouring national visions.

5. Comparative models of quantum-computing education

5.1 What the international workforce and competence literature establishes

Three findings from the international literature bear directly on how Colombia should design its pathway.

First, the demand is real and the required profile is broad. Industry-facing surveys of quantum employers find that companies need not only PhD-level quantum scientists but a much larger population of "quantum-aware" engineers, software developers, and applications specialists who understand enough quantum to be productive without being researchers [19]. The influential "quantum smart workforce" analysis makes the same point from the ecosystem side, arguing for training that spans levels and disciplines rather than concentrating solely on doctoral physicists [20].

Second, the field has begun to codify *which* competencies matter, and the codification is explicitly cross-disciplinary. The European Competence Framework for Quantum Technologies is the most developed instrument: it maps the knowledge, skills, and competence levels a quantum-ready workforce needs, deliberately spanning concepts, hardware, software, and applications [21]. A global review of quantum education initiatives documents how quickly universities, national programs, and industry have moved to build such offerings, and how varied the models are [22]. National coordination bodies have gone further and identified a small set of core concepts that every future quantum-information learner should encounter, providing a shared vocabulary for curriculum designers [23], and validated concept inventories are now emerging to measure whether learners actually acquire those ideas [24].

Third, the pedagogy is maturing, and it points toward simulators, cloud access, and early exposure. Teaching experiences using quantum simulators show that students can build working intuition for the circuit model before they have full physics training [25], and the "second quantum revolution" concept set has been pushed even to the pre-college level, arguing that core quantum-information ideas can and should be introduced early [26]. The vision of quantum communication and a future quantum internet [27] further widens the set of applications a broad workforce should understand.

5.2 Latin American experience

Colombia is not starting in a vacuum regionally. Documented teaching experiences from the region, notably a multi-year effort to bring quantum computing to undergraduate engineering students in Uruguay, show both the appetite and the practical obstacles, and their follow-up work explicitly examines the scope of quantum computing across Latin America [11, 28]. These experiences reinforce the co-teaching lesson: the engineering students could program the devices quickly but needed dedicated scaffolding for the quantum-mechanical and linear-algebra foundations.

5.3 Two national visions: Chile and Mexico

Two neighbouring countries have published national quantum visions that bracket Colombia's approach on the hardware-versus-software axis, and both are instructive comparators. The table below positions all three.

Table 1. National quantum-vision comparison: Colombia (CQA), Chile (2025–2035), and Mexico (IMTC 2023).

Dimension	Colombia (CQA)	Chile (2025–2035)	Mexico (IMTC 2023)
Document type	Community-of-practice roadmap (NGO orchestrator)	State policy / national strategy (Ministry-led)	Scientific-society roadmap (physics community)
Lead actor	CyberColombia + academic/industry partners	Ministerio de Ciencia (with IDB, advisory commission)	DICU — Sociedad Mexicana de Física
Strategic bet	Asset-light, software-first "Algorithm Factory"; hardware via cloud	Full-stack ecosystem: hardware + software + metrology	Hardware/platform-centric (cold atoms, photonics, QKD, sensing)
Education model	Train-the-trainers → quantum master's cohort; literacy first	Full pipeline: schooling → embed in eng/CS/physics/chem → postgraduate specialization	Delocalized "Diplomado" for engineering/physics undergraduates

Dimension	Colombia (CQA)	Chile (2025–2035)	Mexico (IMTC 2023)
Disciplinary framing	Interdisciplinary, use-inspired (Pasteur's quadrant)	Explicit interdisciplinary body (QM + information theory + materials + photonics + cryo-electronics)	Physics-led, platform-anchored; engineering as applier
Sovereignty stance	Sovereignty via application layer and talent, not hardware CapEx	Technological sovereignty as an explicit transversal principle	National capability via own-technology development (QKD, integrated circuits)
Regional integration	Quantum Latino; SCALAC / NCSA cloud partnerships	International-articulation axis; ties to global networks	National-institution network; less explicit regional frame
Time horizon	2026–2036 (three phases)	2025–2035	Short-to-medium term (from 2023)

Chile published, through its Ministry of Science, a *national quantum-technologies strategy for 2025–2035*, a state policy rather than a community roadmap. Its vision is to consolidate a "solid, dynamic, and connected" quantum ecosystem by 2035, and it is notable for two things relevant here. First, it names the interdisciplinary knowledge body explicitly: quantum physics, information theory, advanced-materials engineering, quantum optics and photonics, cryogenic electronics, quantum control, simulation, and high-precision metrology, together with intellectual-property and technology-transfer skills. Second, it lays out a *full talent pipeline* from early schooling (building abstract and mathematical thinking) through technical and university programs that embed quantum content directly into existing engineering, computer-science, physics, chemistry, and applied-mathematics degrees, then postgraduate specialization, and finally insertion of master's and doctoral graduates into science-and-technology companies, supported by teacher training and dual-training schemes. Technological sovereignty is one of six transversal principles, alongside ethics, gender and inclusion, environmental sustainability, transparency, and territorial decentralization.

Mexico's Iniciativa Mexicana en Tecnologías Cuánticas (IMTC, 2023) is different in both authorship and emphasis: it is a roadmap produced by the quantum-information division of the Mexican Physical Society rather than a government policy, and it is decidedly hardware- and platform-centric. Its technical bets are on atomic gravimetry, photonic integrated circuits, non-linear crystals, quantum sensing and imaging, quantum cryptography and key distribution, and entangled-photon-pair sources — leveraging Mexico's strong national base in quantum optics and cold atoms. Its education proposal is correspondingly focused: a *delocalized "Diplomado en tecnologías cuánticas"* aimed at undergraduate students in engineering, physics, and related fields, conceived as an entry door to the quantum job market and distributed across national institutions, with named "education leads" in the initiative's governance.

The contrast is illuminating for Colombia. Mexico bets on hardware platforms grounded in an existing physics-community strength; Chile pursues a comprehensive, government-backed, full-stack strategy with sovereignty as an explicit principle and quantum content embedded across many degrees; Colombia's CQA makes the most sharply focused bet of the three — software and applications only, hardware rented through the cloud. Each bet implies a different educational centre of gravity. Colombia's bet places the centre of gravity precisely on the bridge competencies in the middle of the §3 figure, which is both the strategy's coherence and its risk: if the bridge is not built through deliberate co-teaching, there is no physics-hardware programme to fall back on.

6. A proposed quantum-computing teaching pathway for Colombia

The diagnosis of the preceding sections points to a single design principle: because no existing discipline covers the quantum-computing competency set, and because Colombia's computing tradition is displaced into systems engineering while its quantum expertise is confined to physics, the pathway must be **interdisciplinary by construction and staged by competency**, not a quantum course bolted onto one existing major. We propose a pathway organized in three layers that map directly onto the CQA's 2026–2036 roadmap phases, so that the educational build-out and the ecosystem build-out advance together.

A quantum-computing education pathway for Colombia, aligned to the CQA 2026–2036 roadmap

Layered competencies mapped onto the Colombian Quantum Alliance's three phases — literacy first, then specialization, then sovereign research

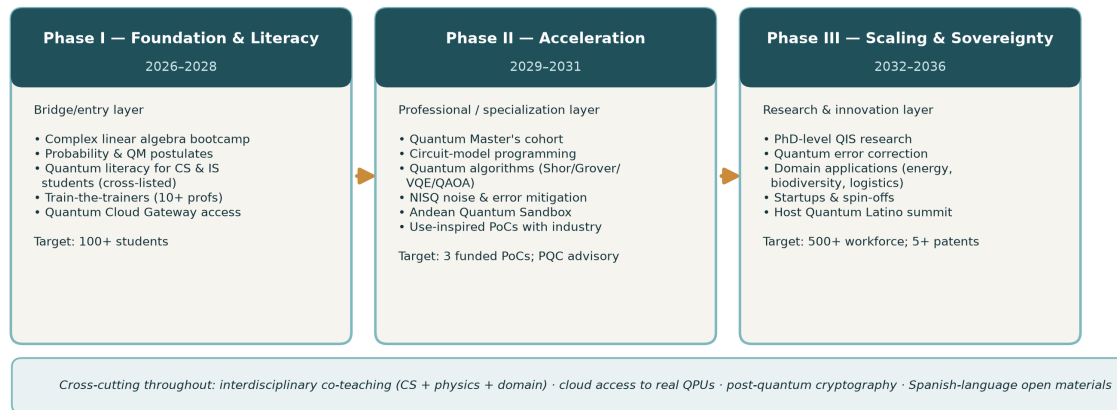


Fig. 3. A proposed quantum-computing education pathway for Colombia, aligned to the Colombian Quantum Alliance's 2026–2036 roadmap: literacy first, then specialization, then sovereign research.

Phase I — Foundation & Literacy (2026–2028): build the bridge deck. The first priority is to close the mathematical and conceptual gaps that §2 identified, for *both* incoming populations. Computer-science and *Ingeniería de Sistemas* students need a complex-linear-algebra bootcamp (Hilbert spaces, tensor products) and a first, honest exposure to the quantum-mechanical postulates, superposition, measurement, entanglement, taught for their computational meaning rather than as a physics survey. Physics and chemistry students need the complementary exposure to programming practice and the circuit model. Both should meet in cross-listed "quantum literacy" courses that use simulators and cloud access to real hardware so that intuition is built through doing, following the evidence that simulator-based and early-exposure teaching works [25, 26]. Aligned with the CQA's train-the-trainers priority and its Quantum Cloud Gateway, this phase should also certify the first cohort of 10+ specialized professors, without them, the later phases have no teachers. The concept vocabulary should be anchored to an established framework (the European Competence Framework and the national core-concepts work) so that "literacy" is defined and assessable rather than vague [21, 23].

Phase II — Acceleration (2029–2031): specialization and the Algorithm Factory. With a literate base and trained faculty, the second layer builds professional competence: a dedicated quantum master's cohort, circuit-model programming in industry-standard frameworks, the canonical quantum algorithms (Shor, Grover, and the variational NISQ-era methods VQE and QAOA), and, crucially for real devices, noise, error mitigation, and an introduction to error correction. This is the layer that produces the CQA's "Algorithm Factory" workforce, and it should be tied to the Andean Quantum Sandbox and to company-funded proofs of concept so that students learn on use-inspired problems rather than toy examples. This directly develops the "domain modelling" competency the matrix marks as weak everywhere, and it matches what the workforce literature says employers actually need, quantum-aware software and applications specialists, not only researchers [19, 20]. The post-quantum-cryptography advisory strand in the CQA roadmap deserves particular emphasis here because it delivers national value *independently of quantum hardware* and is a natural fit for the systems-and-security strengths of the existing *Ingeniería de Sistemas* base.

Phase III — Scaling & Sovereignty (2032–2036): research, applications, and spin-offs. The third layer supports PhD-level quantum-information research, deeper quantum error correction, and domain-application programs in nationally relevant areas (energy-grid optimization, biodiversity, logistics). It is also where the workforce reaches scale (the CQA's 500+ target) and where startups and the first sovereign patents emerge, supported by hosting the regional Quantum Latino summit. Sovereignty here is understood in the CQA's asset-light sense, sovereignty of talent, algorithms, and applications rather than of fabrication, which is the realistic form of sovereignty for Colombia's constraints.

Cross-cutting throughout. Four commitments run across all three phases and are, in our view, non-negotiable given the diagnosis. (1) *Interdisciplinary co-teaching*: every course should be delivered by, or jointly designed across, computer science, physics, and a domain area, because the interdisciplinary classroom is the documented reality of

quantum education [13, 14]. (2) *Cloud access to real QPUs* from day one, consistent with the asset-light strategy. (3) *Post-quantum cryptography* as a persistent applied strand that pays off regardless of hardware timelines. (4) *Spanish-language open materials*, both to widen access nationally and to serve the Quantum Latino regional agenda. Chile's strategy embeds quantum content across many existing degrees and Mexico's delivers a delocalized diploma; Colombia's version of "embedding" is this cross-listed, co-taught, cloud-based bridge that connects its displaced computing tradition to its narrow quantum one.

6.1 How the pathway maps to the competency matrix

The pathway is designed so that, by its end, a graduate has covered every row of the §2 matrix that no single discipline covers today. Phase I closes the complex-linear-algebra and QM-postulate gaps and begins quantum information theory and the circuit model. Phase II closes quantum algorithms, amplitude/interference reasoning, and error/noise handling while adding domain modelling and cloud/platform engineering. Phase III deepens error correction and research competence. The classical software-engineering and computational-thinking strengths that CS and *Ingeniería de Sistemas* already provide are retained as the base the whole structure stands on — the pathway *adds* the quantum intersection rather than replacing the computing core.

7. Conclusion

Traditional computer-science education is a necessary but insufficient foundation for quantum computing, and the insufficiency is structural: the competencies that define quantum computing sit at an intersection that computer science, physics, and — in Colombia — *Ingeniería de Sistemas* each cover only in part. The competency matrix makes the shape of the gap explicit; the six quantum-specific competencies are absent from both the computer-science and the systems-engineering curricula, present but partial in physics, and covered fully by no one. Colombia experiences this gap with unusual force because its computing discipline is young and displaced into systems engineering, while its quantum expertise is concentrated in a single physics-based program characterized in the literature as research without applications [7].

The Colombian Quantum Alliance's asset-light, software-first strategy is well matched to these constraints, but it raises the educational stakes rather than lowering them: an application-and-algorithm bet with no hardware fallback depends entirely on building the bridge competencies through deliberate, interdisciplinary co-teaching [8]. The international literature converges on how to do this — a broad quantum-aware workforce, a codified cross-disciplinary competency framework, simulator- and cloud-based pedagogy, and early literacy, and the neighbouring national visions of Chile (full-stack, state-led, sovereignty-as-principle) and Mexico (physics-community, hardware-platform) show the range of alternatives Colombia is choosing among. The phased pathway proposed here translates that evidence into a build-out aligned to the CQA's own 2026–2036 timeline: literacy and trained trainers first, specialization and the Algorithm Factory second, research and scaled sovereignty third, with interdisciplinary co-teaching, cloud access, post-quantum cryptography, and Spanish-language open materials running throughout. The central recommendation is simple to state and hard to execute: Colombia should not try to make its computer scientists into physicists or its physicists into software engineers, but should build the institutional bridge, co-taught, cloud-based, and use-inspired, on which both can meet.

Note on scope and sources

This is a working review intended for iteration. Its empirical claims about the Colombian ecosystem draw primarily on the triple-helix analysis of [7] and on the Colombian Quantum Alliance's published vision; the Chilean and Mexican national visions are characterized from the strategy documents supplied for this analysis. The competency matrix reflects typical program structures (ACM/IEEE-style computer science, physics-with-quantum-mechanics, and ACOFI-style *Ingeniería de Sistemas*) rather than any single institution's curriculum, and is offered as an analytical instrument to be refined against specific Colombian program data.

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